

Azure DevOps

Training / Learning Documentation

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◇ What is DevOps?

- **DevOps = Development + Operations.** It's a way of working where software developers (who write code) and operations teams (who run and manage systems) **work together closely** instead of separately.

◇ Why DevOps exists

- In the past, developers built software and then “threw it over the wall” to operations to run it. This caused delays, bugs, and blame games.
- DevOps removes that wall. It makes both teams **share responsibility** for building, testing, deploying, and maintaining software.

◇ Key Ideas in DevOps

- **Automation** → Repetitive tasks (like testing, building, deploying) are done by tools, not humans.
- **Continuous Integration (CI)** → Developers merge code often, and it's automatically tested.
- **Continuous Delivery/Deployment (CD)** → Software updates are automatically pushed to production quickly and safely.
- **Monitoring** → Systems are watched in real time to catch problems early.
- **Collaboration** → Developers and operations use shared tools and processes.

DevOps is a culture + set of practices + tools that help teams deliver software faster, with fewer errors, and with better teamwork.

◇ Advantages of DevOps

1. **Faster Delivery of Software**
 - a. Code changes move quickly from development to production.
 - b. This means new features, bug fixes, and updates reach users faster.
2. **Better Collaboration**
 - a. Developers and operations teams work together instead of separately.
 - b. This reduces misunderstandings and improves teamwork.
3. **Higher Quality Software**
 - a. Continuous testing and monitoring catch errors early.
 - b. Less chance of big failures in production.

4. Automation Saves Time

- a. Repetitive tasks (like testing, building, deploying) are automated.
- b. Teams spend more time on innovation instead of manual work.

5. Scalability & Reliability

- a. Systems can handle more users and traffic smoothly.
- b. Monitoring ensures problems are detected and fixed quickly.

6. Continuous Improvement

- a. Feedback loops (from monitoring and users) help teams improve constantly.
- b. Software evolves steadily instead of waiting for big releases.

7. Reduced Costs

- a. Faster fixes and fewer failures mean less wasted money.
- b. Automation reduces the need for manual effort.

DevOps makes software delivery faster, safer, cheaper, and more team-friendly.

◇ What is Agile?

- Agile is a way of developing software step by step, in small parts, instead of doing everything at once.
- It focuses on **flexibility, teamwork, and quick delivery** of working software.

◇ Why Agile?

- In traditional methods (like Waterfall), teams planned everything upfront and delivered only at the end. If requirements changed, it was hard to adapt.
- Agile solves this by working in **short cycles (called sprints)**, so teams can adjust quickly to changes.

◇ Key Principles of Agile

1. **Customer Collaboration** → Work closely with customers, get feedback often.
2. **Working Software First** → Deliver small, usable features quickly instead of waiting months.
3. **Responding to Change** → If requirements change, adapt easily.
4. **Teamwork & Communication** → Developers, testers, and business people work together daily.

◇ Simple Example

Imagine you're building an e-commerce app:

- In **Sprint 1**, you build login and signup.
- In **Sprint 2**, you add product browsing.
- In **Sprint 3**, you add checkout. At the end of each sprint, the customer can **see and use** the new feature, give feedback, and you improve it.

Agile = flexible, fast, customer-focused way of building software in small steps, with constant feedback and teamwork.

◇ What is Scrum?

- **Scrum is a way of working inside Agile.**
- It's a **framework** that helps teams organize their work, deliver software in small steps, and improve continuously.
- Scrum is a structured way of doing Agile, where teams work in short cycles (Sprints), guided by clear roles and meetings, to deliver software step by step with constant feedback.
- it gives a structure for doing Agile using things like sprints, product backlog, daily standups, Product Owner, and Scrum Master.

◇ How Scrum Works

- Work is divided into **short cycles called Sprints** (usually 2–4 weeks).
- At the end of each Sprint, the team delivers a **usable piece of software**.
- The team meets regularly to plan, review, and improve.

◇ Key Roles in Scrum

1. **Product Owner** → Decides what features are most important, manages the product backlog (list of tasks).
2. **Scrum Master** → Helps the team follow Scrum rules, removes obstacles, acts like a coach.
3. **Development Team** → Builds the product, self-organizes, and decides how to do the work.

◇ Scrum Events (Meetings)

- **Sprint Planning** → Decide what to build in the next Sprint.
- **Daily Stand-up** → Quick 15-min meeting each day to sync progress.
- **Sprint Review** → Show the finished work to stakeholders and get feedback.
- **Sprint Retrospective** → Team reflects on what went well and what can improve.

◇ Simple Example

Imagine you're building a shopping app:

- Sprint 1 → Add login/signup.
- Sprint 2 → Add product browsing.
- Sprint 3 → Add checkout. After each Sprint, the app grows step by step, and users can test features early.

Scrum is a structured way of doing Agile, where teams work in short cycles (Sprints), guided by clear roles and meetings, to deliver software step by step with constant feedback.

◇ What is a Product Owner (PO)?

- The **Product Owner** is a key role in **Scrum**.
- They are the **person responsible for deciding what the team should build** and in what order.

◇ Responsibilities of a PO

1. **Manages the Product Backlog**
 - a. The backlog is a list of all features, fixes, and tasks.
 - b. The PO organizes it by priority (what's most important first).
2. **Represents the Customer/Business**
 - a. The PO speaks for the customer and stakeholders.
 - b. They make sure the team builds things that bring the most value.
3. **Defines Goals for Each Sprint**
 - a. The PO explains what needs to be done in the upcoming Sprint.
 - b. They set clear objectives so the team knows the target.
4. **Accepts or Rejects Work**
 - a. At the end of a Sprint, the PO reviews the work.
 - b. If it meets the requirements, they accept it; if not, they ask for changes.

◇ Simple Example

Imagine you're building a shopping app:

- The **PO** decides: "Login is more important than checkout right now."
- They put **Login** at the top of the backlog.
- The team builds it in Sprint 1.
- The PO checks it, gives feedback, and then moves on to the next priority (like product browsing).

The Product Owner is the "decision-maker" in Scrum who prioritizes work, represents the customer, and ensures the team builds the right product step by step.

◇ What is a Scrum Master?

- The **Scrum Master** is the **coach and guide** for the Scrum team.
- They don't manage the team like a boss — instead, they **help the team follow Scrum rules, remove obstacles, and work smoothly together**.

◇ Responsibilities of a Scrum Master

1. Facilitates Scrum Events

- a. Makes sure meetings like Sprint Planning, Daily Stand-ups, Reviews, and Retrospectives happen properly.

2. Removes Roadblocks

- a. If the team faces issues (like missing tools, unclear requirements, or external blockers), the Scrum Master helps solve them.

3. Protects the Team

- a. Shields the team from distractions or unnecessary interruptions so they can focus on delivering work.

4. Teaches & Coaches

- a. Guides the team and organization in understanding Agile and Scrum principles.
- b. Helps improve teamwork and processes.

5. Encourages Self-Organization

- a. The team decides how to do the work; the Scrum Master ensures they have the environment to succeed.

◇ Simple Example

Imagine you're building a shopping app:

- The **Product Owner (PO)** says: "Login is the top priority."
- The **Development Team** builds it.
- The **Scrum Master** makes sure the Sprint runs smoothly, meetings happen, and if the team is stuck (say, waiting for server access), they fix that issue.

The Scrum Master is like a coach or facilitator who ensures the team follows Scrum, removes obstacles, and helps them work effectively — not a boss, but a guide.

The Product Owner is the decision-maker for product features, while the Scrum Master is the coach who ensures the team follows Scrum properly and works effectively

◇ Advantages of Agile

1. Faster Delivery

- a. Work is done in small cycles (sprints), so usable features reach customers quickly instead of waiting months.

2. Flexibility to Change

- a. If customer needs or market conditions change, Agile teams can adapt easily without wasting effort.

3. Better Quality

- a. Continuous testing and feedback during each sprint catch problems early, leading to fewer big failures.

4. Customer Satisfaction

- a. Customers see progress regularly, give feedback, and feel involved in shaping the product.
- 5. Improved Team Collaboration**
 - a. Developers, testers, and business people work closely together, reducing misunderstandings and delays.
- 6. Transparency**
 - a. Daily stand-ups and sprint reviews make progress visible to everyone, so there are no surprises.
- 7. Risk Reduction**
 - a. Delivering in small increments lowers the risk of project failure because issues are spotted early.
- 8. Continuous Improvement**
 - a. Retrospectives after each sprint help the team learn from mistakes and improve their process.

◆ Simple Analogy

Agile is like **building a house room by room**:

- Instead of waiting until the whole house is finished, you complete one room, show it to the owner, get feedback, and then move to the next.
- This way, the owner gets value step by step, and changes can be made easily before the whole house is done.

Agile makes software delivery faster, more flexible, customer-focused, and less risky, while improving teamwork and quality.